

2024



Goodnotes Pens the Future by Developing Generative AI Features

Discover how Goodnotes enhanced its generative AI capabilities on Anthropic Claude on Amazon Bedrock in just a few months, facilitating rapid scalability.

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Accelerated

go-to-market for new AI-powered features. [Ask Goodnotes](#)



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of active users supported, with high service quality



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Innovation

Provides reliable foundation for rapid innovation and flexibility for future LLMs



Safeguards users

Maintains high data privacy and security standards

Overview

[Goodnotes](#), known for its digital note-taking and productivity app, aimed to enhance its generative AI capabilities by refining its AI infrastructure to improve scalability and cost-effectiveness. This ensures the platform can handle growing user demand while maintaining high performance and keeping operational costs low.

In 2024, Goodnotes transitioned from a self-hosted model on [Amazon Elastic Kubernetes Service](#) (Amazon EKS) to [Anthropic Claude](#) on [Amazon Bedrock](#) to support the launch of its AI-powered Ask Goodnotes feature. This feature empowers users to leverage generative AI to explore, learn from, and build on the knowledge and ideas captured in their Goodnotes notebooks. This fully managed service helped Goodnotes accelerate its go-to-market efforts and minimize the time and resources required for infrastructure management, allowing the team to focus on advancing AI development, while ensuring rapid scalability to meet increasing user demand.



Opportunity | Transforming the Note-Taking Experience with Generative AI

Founded in 2011, Goodnotes has been a leader in digital note-taking and productivity, now serving over 24 million monthly active users. As a pioneer in digital handwriting, Goodnotes was among the first apps to support the Apple Pencil when it launched in 2015. Since then, it has become an essential tool, earning recognition as Apple's iPad App of the Year in 2022. In recent years, Goodnotes has evolved into a comprehensive platform for productivity, creativity, and learning, offering features like text typing, PDF annotation, commenting, collaboration, and more.



Goodnotes has invested significantly in AI and ML to improve user efficiency and enhance the experience of work and study. In August 2023, the company introduced AI-powered handwriting enhancements, becoming the first app to offer spell check and word completion in users' own handwriting styles. These innovations align with Goodnotes' broader mission to empower users with AI tools that accelerate productivity and learning.

To support some of these new features, initially, Goodnotes adopted a large language model (LLM) on Amazon Elastic Kubernetes Services (Amazon EKS), leveraging a GPU-based cluster for scalable, cost-effective performance. This infrastructure, combined with [Amazon Elastic Compute Cloud](#) (Amazon EC2) instances and [Elastic Load Balancing](#), reduced latency and cut costs by 97 percent compared to SaaS-based LLMs.

In 2024, Goodnotes focused on streamlining its AI infrastructure and accelerating the development of generative AI features, including Ask Goodnotes, an AI-powered assistant. Ask Goodnotes is designed to help users unlock the full potential of their ideas and information. Drawing directly from users' content—typed text, handwritten notes, or imported PDFs—it answers questions, provides summaries and explanations, extracts insights, and even generates custom quizzes.

In addition to providing these new capabilities right within the app, Goodnotes sought to refine its AI infrastructure for increased scalability and cost-effectiveness, ensuring the platform could handle growing user demand while maintaining performance and keeping operational costs sustainable.

Solution | Optimizing AI Infrastructure with Anthropic Claude on Amazon Bedrock

To deliver its generative AI capabilities at scale and streamline its development process, Goodnotes transitioned to using Amazon Bedrock to access selected LLMs. Amazon Web Services (AWS) provides access to a wide range of industry-leading companies like Anthropic and Cohere through Amazon Bedrock, offering customers ongoing flexibility to try different LLMs.

Amazon Bedrock provides Goodnotes with a fully managed service that delivers scalability, flexibility, and security, allowing the company to efficiently manage large volumes of data and user requests. It also enables Goodnotes to



experiment with and deploy multiple LLMs without the need for extensive infrastructure management, leading to faster innovation cycles and reduced overhead. “With Amazon Bedrock, we have streamlined AI feature delivery, accelerated innovation and iteration, and focused more on optimizing the user experience,” explains Angus Fong, AI team lead and engineering manager at Goodnotes.

Additionally, Amazon Bedrock’s guardrail capabilities ensure that responses generated by LLMs are accurate, reliable, and aligned with the needs of Goodnotes’ users. These guardrails enhance the quality of AI responses and improve the learning and productivity experience for users who rely on Goodnotes.

Building on AWS also helps ensure that Goodnotes’ customer data is securely protected. Nebahat Arslan, group general counsel at Goodnotes, mentions, “The experience we’ve had working with AWS in upholding our data privacy and security standards gave us confidence to develop further generative AI features with their services. For example, as part of our commitment to empowering users with complete control over their information, we’re dedicated to never collecting user data without explicit consent—and we needed infrastructure with robust safeguards and data protections.”

Goodnotes utilized AWS security services, including [Amazon CloudWatch](#), [Amazon GuardDuty](#), and [Audit Manager](#), as part of its preparation for ISO 27001 certification, which it achieved in November 2024. After a successful waitlist preview, Goodnotes anticipated strong demand for Ask Goodnotes and rapidly scaled the feature to millions of users. AWS supported this launch by monitoring performance and offering development frameworks that accelerated deployment. This allowed Goodnotes to focus on refining Ask Goodnotes, delivering a smooth experience to its expanding user base. Additionally, with AWS’ support, Goodnotes transitioned to a cross-region inference architecture, minimizing risks tied to single host locations and improving service availability. This close collaboration helped Goodnotes maintain its high standards while iterating quickly on feature enhancements.

Goodnotes prioritizes delivering reliable and consistent service to its users, supported by technical expertise from AWS. “When users depend on us as an essential day-to-day tool for their work, studies, or personal pursuits, it’s crucial to maintain reliability and availability, and in turn for us to get quality technical support from our service providers,” explains Fong. Even after opening access to millions of users, Goodnotes was able to achieve high reliability for its main API service supporting Ask Goodnotes.

Goodnotes reinforced this level of reliability through proactive collaboration with AWS, focusing on identifying, investigating, and resolving potential risks to ensure consistent service.

Outcome | Accelerating Go to Market and Scaling to Millions of Users

By utilizing the performance, scalability, and cost-effectiveness of running its LLM on AWS, Goodnotes has been able to speed up the development of new generative AI features for its app. These advancements have enabled Goodnotes to quickly introduce new functionalities, ensuring that users can access the latest AI innovations directly within the app, where they are already capturing and reviewing their notes and information.

Since migrating to LLMs on Amazon Bedrock, the company has been able to support millions of users simultaneously, ensuring a smooth experience even during peak usage times. “We accelerated our go-to-market efforts, reducing development time and confidently launching Ask Goodnotes for millions of users simultaneously. This was achieved while enabling our team to focus on further development and user experience, all supported by AWS’s expertise in infrastructure management and AI model maintenance,” says



Looking ahead, Goodnotes is committed to creating more AI-powered features tailored to its diverse user base, including office and field professionals, students and teachers, precise planners, and creative doodlers. Steven Chan, founder and CEO at Goodnotes, adds, “Working with AWS enables us to provide Goodnotes users with the best LLM-powered experience by allowing rapid testing of various models during development and maintaining the flexibility to experiment with and implement cutting-edge models as they emerge.”

About Goodnotes

Goodnotes is a widely used AI-powered digital paper platform, trusted by millions of users worldwide. Since its launch in 2011, Goodnotes has transformed note-taking by combining the familiarity of handwriting with the convenience of digital organization. Users can take handwritten digital notes, search handwritten text, and keep their work organized in a digital library. Today, Goodnotes continues to innovate with generative AI for digital handwriting in the productivity space. It was recognized as Apple’s iPad App of the Year in 2022.

AWS Services Used

Amazon Bedrock

The easiest way to build and scale generative AI applications with foundation models

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Amazon Elastic Kubernetes Service

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